AgentOps

THE AGENTIC EDITION

IDEA SUBMISSION

AdmitSense Admission Agent



Helps Families prepare an insurance review packet that updates when evidence changes

9to5 Victims · HX-Q97Y

Track 03: Tool-using and RAG agents

MIT Bengaluru · 24 October 2026

School of Computer Engineering & SDG Club × CGI

The problem worth solving

Families must reconcile policy conditions and missing documents while arranging hospital admission.

WHO EXPERIENCES IT?

Patients and caregivers preparing for admission, supported by the hospital insurance desk.

WHAT HAPPENS TODAY?

Read policy PDFs, collect records, and contact the insurer. New facts can require another review.

WHY IT MATTERS

Insurer review follows hospital submission of a pre-authorisation request [1]. Hypothesis: a clearer evidence packet reduces avoidable follow-up.

Your proposed solution

AdmitSense investigates an admission case and prepares a cited assessment with the next unresolved question.

01 USER INPUT

Policy PDFs and endorsements.
Procedure and hospital.
Synthetic patient details

02 AGENT ACTION

Find clauses and missing facts.
Check hospital records.
Revise affected findings

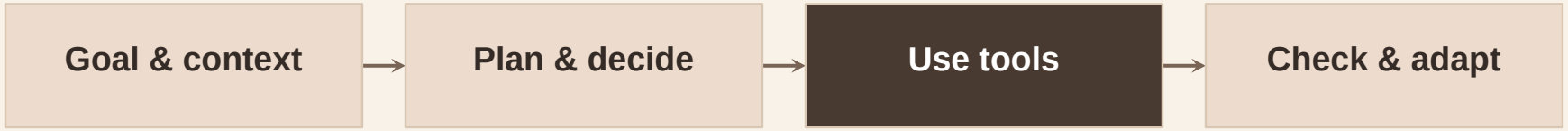
03 USER OUTCOME

Cited findings and a checklist.
Questions for the insurer.
Separate authorisation status. Faster Decisions

CORE VALUE Evidence Change Replay explains each revision and the next action.

How the agent works

Goal: resolve evidence gaps. Choose which clause, document or question to investigate next.



Goal & context	Plan & decide	Use tools	Check & adapt
Case details and versioned policy text. No assumed approval	Choose a missing fact that could change the assessment.	Retrieve pages, check hospital records, update the packet.	Check citations. New evidence triggers a revised plan.

HUMAN CONTROL & STOP RULES

Pause on missing facts or conflicts. Limit: 8 tool steps. User reviews drafts. Insurer decides authorisation.

Technology & data

DATA & INPUTS

One policy product in text PDFs.
Synthetic schedule and patient cases.
Page-indexed clause retrieval.
Labelled hospital dataset with dates.

MODELS & ORCHESTRATION

Python + Gemini API / groq.
One orchestrator with specialist roles.
Pydantic validates output structure.
Python checks dates and evidence IDs.

TOOLS & INTEGRATIONS

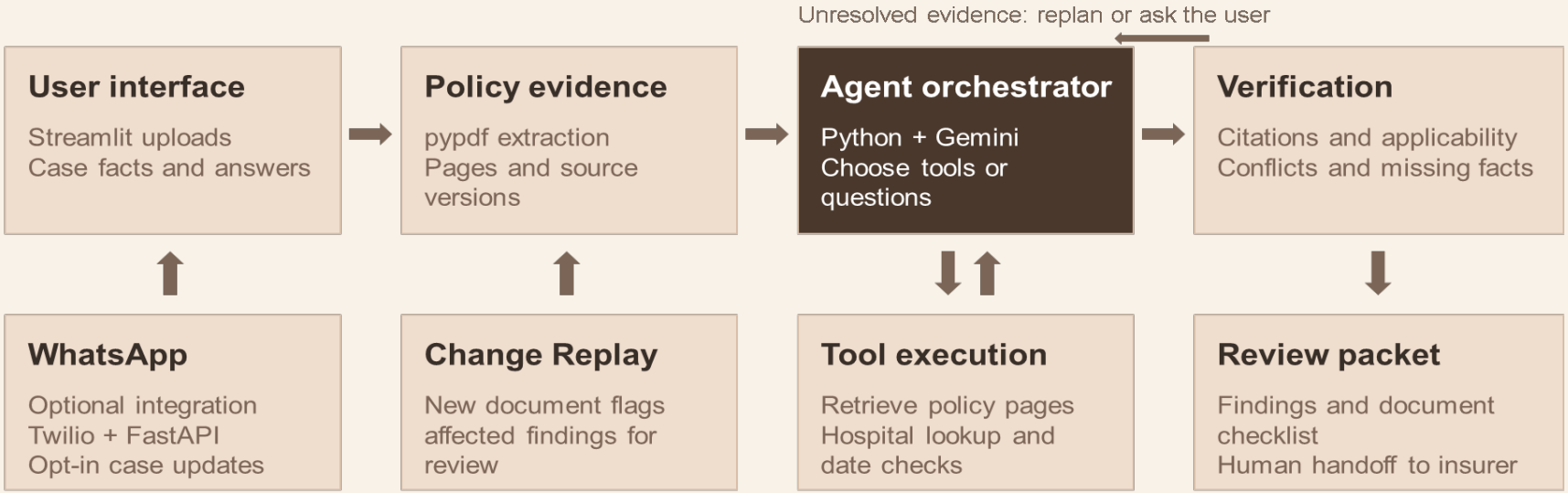
Streamlit interface and session state.
pypdf extracts text with page numbers.
JSON stores hospital records.
HTML export creates the review packet.

DEPENDENCIES & LIMITS

API key, quota and latency to verify.
No live insurer integration or OCR.
WhatsApp stretch: Twilio + webhook.
Account and delivery access pending.

Insurance Admission Agent

New evidence changes the investigation, the findings and the next action.



Storage: SQLite case history + local documents. Hospital lookup uses labelled demo data.

Policy assessment, hospital-network evidence and insurer authorisation remain separate.

What makes it different

CURRENT APPROACH

Policy explanations and claim-document assistance [2, 3]

PolicyGPT explains cover. ClaimGPT describes document checks and reports. Our focus: make revisions traceable.

YOUR APPROACH

Evidence Change Replay for admission readiness

Link findings to source versions. Flag affected findings after an update. Ask the next question and rebuild the packet.

TRACK & RELEVANCE

03 / Tool-using and RAG agents
Evidence guides tools, follow-up questions and revisions.

A new document changes the plan

BEFORE THE UPDATE

Sample policy: 24-month waiting period.

Sample policy age: 18 months.

Finding: waiting condition unresolved.

Agent requests relevant endorsements.

Hospital status stays a separate finding.

Fictional policy conditions and patient.
No real coverage decision or insurer approval.

AFTER THE UPDATE

Endorsement Arrives

Policy ID and effective date match.

It changes this waiting period to 12 months.
Agent supersedes the affected finding.

Next: request the missing hospital estimate.Rebuild the checklist and review packet.

Authorisation status: pending insurer review

What you will build

MINIMUM WORKING VERSION

One policy product, three synthetic cases, a hospital lookup and a review packet.
The demo adds an endorsement and shows an evidence-driven revision.

BUILD

PDF extraction and interface.
Owner: UI/data role

INTEGRATE

Tools and change tracking.
Owner: agent role.
Local hospital lookup.

VALIDATE

Test, fix and rehearse.
Check normal, ambiguous and unreadable-input cases.

IF TIME ALLOWS

WhatsApp companion: opt-in checklist alerts through Twilio Sandbox, giving live feedback during emergencies for quicker and efficient decision making.

Impact, evaluation & risks

EXPECTED BENEFIT

Help caregivers organise the next step and give insurance desks a traceable summary to review.

SUCCESS MEASURE

Targets: 3/3 expected case outcomes.
All cited pages resolve correctly.
Zero unsupported approval claims.

KEY RISK OR FAILURE CASE

A missed exception or an incorrect source priority changes the assessment.
Uploaded PDFs may contain malicious instructions or sensitive information.

SAFEGUARD & FALLBACK

Check policy identity and effective dates.
Treat document text only as evidence.
Flag conflicts for human review.
Use synthetic data and server-side keys.

Planned tests, not measured results. The tool supports administration and must not delay emergency care.

The team behind the idea

TEAM LEADER

Dev Srijit · MIT · 4th year
Role: Agent Logic and Evaluation

SECOND MEMBER, IF APPLICABLE

Dhruv Srijit · MIT · 4th year
Role: Interface and Data

REFERENCES & ACKNOWLEDGEMENTS

[1] Niva Bupa: cashless claim workflow. [2] Plum: Introducing PolicyGPT.
[3] ClaimGPT: public project description. [4] Gemini, Streamlit and pypdf docs.

Fully Built, Integrated and Verified MVP

Transformed from an idea into an end-to-end working system with live LLM execution and regression testing.

CORE SYSTEM DELIVERY

End-to-end workflow

• Live Streamlit UI for case intake, PDF upload, review, Evidence Change Replay and HTML packet export.
• SQLite persistence with immutable versioned audit ledger using CaseRepository and ReviewRepository.
• Deterministic tool gatekeeper for page search, physical PDF reading and synthetic hospital lookup.

LIVE LLM PIPELINE

Groq at runtime

• Provider and model: Groq with openai/gpt-oss-120b and sub-second response latency.
• Structured semantic extraction with JSON schema validation.
• Zero silent mocks: telemetry discloses model, token usage and execution mode.

TEST & QUALITY MATRIX

Verified delivery

• 152 / 152 automated tests passing across unit, retrieval grounding and concurrency.
• 24 / 24 milestone verification checks passed across M1 through M7.
• 3 validated scenarios: applicable endorsement, mismatched policy ID and unreadable PDF.

PRODUCTION STATUS **Deployed locally on Streamlit with zero mock dependencies for review replay.**

Safety-First Architecture & Hallucination Defense

Strict guardrails ensure AdmitSense acts as an administrative assistant, never assuming unverified authority.

01 ZERO-HALLUCINATION CITATION GROUNDING

• Verbatim substring rule: every cited clause must match an exact physical substring from the uploaded PDF page.
• Rejection on failure: altered or hallucinated text is physically invalidated by code.

02 NON-NEGOTIABLE AUTHORIZATION INVARIANT

• Hardcoded invariant: authorization_status defaults to and remains “not verified”.
• AdmitSense never fakes insurer approval, cashless pre-authorization or clinical eligibility. Human TPA decides.

03 BOUNDED AGENT GUARDRAILS

• Hard eight-tool-call cap blocks call #9 to prevent runaway costs or infinite loops.
• Deterministic timeouts halt execution gracefully while preserving partial findings.

04 SECURITY & PROMPT INJECTION DEFENSE

• Uploaded documents remain passive, read-only evidence. Policy text cannot override system rules.
• Strict tenant scoping prevents cross-case document queries or leakage.

RESPONSIBILITY **No automated claims settlement without human TPA oversight.**

Measurable Impact and Extensible Architecture

Eliminating hospital admission friction while offering an enterprise-ready, modular foundation.

BUSINESS VALUE & ROI

Lower friction at admission

70% — **Faster admission preparation**

Removes manual review of 50+ page policy PDFs at hospital desks.

PRE-AUTH — **Fewer avoidable rejections**

Catches waiting periods, sub-limits and missing continuity documents before arrival.

CAREGIVER — **Clearer next questions**

Gives families precise, cited questions for the hospital TPA desk.

<$0.02 — **Per review**

Bounded tool execution and targeted page retrieval keep review costs low.

REUSABILITY & EXTENSIBILITY

A modular foundation

PROVIDER-AGNOSTIC

GroqReviewClient uses standard JSON contracts, so the model can be swapped for Gemini, Claude or local Ollama without changing agent logic.

MODULAR SERVICES

Retrieval, hospital directory, review engine and storage remain independent services.

STANDARDIZED SCHEMAS

Pydantic models and JSON export are ready for hospital HIS and FHIR / ABDM integration.

DESIGNED FOR **Hospital TPA desks, patient advocacy portals and health-tech admission engines.**